Vanishing center-of-mass diffusion for the three-dimensional FENE dumbbell model

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Abstract

We study the zero center-of-mass diffusion limit for the full kinetic FENE dumbbell model on the three-dimensional torus. For every solvent viscosity $\nu > 0$, FENE exponent $k > 2$, smooth divergence-free initial velocity, and smooth normalized initial density with bounded positive Maxwellian quotient, we select global positive-diffusion weak solutions converging to a global weak solution of the identical nondiffusive system. Velocity converges strongly in L^2 and polymer density and stress converge strongly in L^1 on finite time intervals. All transport and configuration fluxes have their original constitutive forms in the limit. The proof conditions on a spatial Brownian motion common to every polymer configuration. Nonlinear configuration entropies are transformed before expectation, giving only a linear spatial diffusion term. Conditional stress cutoffs, a transported limiting normalizer, and a joint covariance estimate then propagate a zero entropy defect without assuming strong convergence of the velocity gradient.

Introduction

In a kinetic dumbbell model, the fluid transports a probability distribution of polymer configurations, and the corresponding elastic stress feeds back into the fluid momentum equation. Finitely extensible nonlinear elasticity confines the configuration to a ball by a potential singular at its boundary. Center-of-mass diffusion describes translational Brownian motion of the polymer molecules. Removing this term should preserve both the distribution and its coupling to the full velocity gradient. For weak solutions at large data, this requires a compactness assertion beyond separate existence for the diffusive and nondiffusive equations.

Barrett and Süli [1] established large-data weak existence with positive center-of-mass diffusion. Masmoudi [2] established weak existence without that diffusion by propagating a transformed-density defect and using stress-driven Stokes dissipation. The distinction between those existence results and passage from one model to the other is explicitly described in [4, Section 1]. The result below addresses that passage for the full noncorotational model on $\mathbb{T}^3 \times B_1$, with arbitrary prescribed data in the stated smooth bounded-positive class. No smallness condition is imposed.

The elementary estimates already give strong velocity compactness and identify the singular stress using a configuration boundary tail. They do not identify the product of a weakly convergent velocity gradient with a weakly convergent configuration moment. The nondiffusive entropy-defect argument provides a way to close this correlation, but applying its nonlinear entropy quotients directly to a spatially diffusive density creates spatial Hessian terms whose sign is unavailable.

We instead represent each positive-diffusion density by conditioning on one spatial Brownian motion shared by all labels and configurations. Given this motion, each spatial

characteristic carries an ordinary configuration Fokker–Planck equation. Its entropy and logarithmic moments can be computed before expectation. The expected transformed equation then has just a linear spatial Laplacian, which disappears against fixed tests as its coefficient tends to zero. A fixed-velocity comparison identifies this representation with the chosen coupled density.

The remaining argument keeps this conditional structure when removing entropy cutoffs. Radial positivity preserves the monotone stress squares, and the deterministic velocity gradient gives a common direction to the cutoff drags. These facts identify the stress-work covariance through a local Stokes energy comparison. A joint law of the gradient, conditional square root, and configuration derivative retains the positive Fisher moments. An explicit two-state inequality supplies their averaged coercivity. Finally, a normalizer transported by the limiting fluid makes the entropy defect bounded and gives a characteristic Gronwall inequality. This is an extension of the defect method in [2]; the conditional construction and its averaged cutoff and coercivity arguments are proved here rather than attributed to that work.

Section 1 states the precise result. Sections 2 and 3 provide the uniform estimates and identify the conditional representation. Sections 4–6 establish the common normalizer, stress-work identity, and configuration coercivity. The normalized defect closes in Section 7, and Section 8 passes the original system and its initial-time entropy inequality to the limit.

1 The system and the compactness theorem

Let $B = B_1(0) \subset \mathbb{R}^3$ and let $\mathbb{T}^3$ have its usual periodic measure. Fix $\nu > 0$ and $k > 2$, and set

$$U(q) = -k \log(1 - |q|^2), \qquad M(q) = Z^{-1}(1 - |q|^2)^k, \qquad Z = \int_B (1 - |q|^2)^k \, \mathrm{d}q. \tag{1.1}$$

We consider the full velocity-gradient FENE system

$$\partial_t u + u \cdot \nabla_x u - \nu \Delta_x u + \nabla_x p = \operatorname{div}_x \tau(\psi), \qquad \operatorname{div}_x u = 0, \tag{1.2}$$

$$\tau(\psi) = \int_B K(q)\psi \, \mathrm{d}q, \qquad K(q) = q \otimes \nabla_q U(q). \tag{1.3}$$

$$\begin{aligned} \partial_t \psi + u \cdot \nabla_x \psi &= \varepsilon \Delta_x \psi \\ &\quad + \operatorname{div}_q\{M \nabla_q(\psi/M) - (\nabla_x u) q \psi\}. \end{aligned} \tag{1.4}$$

The configuration flux has zero normal component at ∂B. Its weak meaning is the variational form of (1.4) for smooth tests on the closed ball, with no configuration boundary term. Thus no value of ψ/M on the sphere is prescribed. We use $(\nabla_x u)_{ij} = \partial_{x_j} u_i$; radial symmetry makes K symmetric.

Write $L^p_M = L^p(B, M \, \mathrm{d}q)$ and $H^1_M = \{h : \int_B M(h^2 + |\nabla_q h|^2) < \infty\}$. For a normalized density define

$$\begin{aligned} \operatorname{Ent}_M(\psi) &= \int_{\mathbb{T}^3 \times B} \psi \log(\psi/M), \qquad \mathcal{E}(u, \psi) = \tfrac{1}{2}\|u\|_2^2 + \operatorname{Ent}_M(\psi), \\ I_q(\psi) &= \int_{\mathbb{T}^3 \times B} M |\nabla_q \sqrt{\psi/M}|^2. \end{aligned} \tag{1.5}$$

The spatial Fisher information I_x is defined by replacing ∇_q by ∇_x. The usual initial-time energy inequality is

$$\mathcal{E}(u(t), \psi(t)) + \nu \int_0^t \|\nabla_x u\|_2^2 \, \mathrm{d}s + 4 \int_0^t I_q(\psi) \, \mathrm{d}s + 4\varepsilon \int_0^t I_x(\psi) \, \mathrm{d}s \le \mathcal{E}(u_0, \psi_0). \tag{1.6}$$

All assertions about this inequality at a positive time are understood for almost every time, with the usual weakly continuous representatives.

Theorem 1.1. *Let u_0 be a smooth divergence-free periodic velocity, and let $\psi_0 = Mg_0$, where g_0 is smooth on $\mathbb{T}^3 \times B$, bounded above and below by positive constants, and $\int_B Mg_0(x,q)\,\mathrm{d}q = 1$ for every x. For every $\nu > 0$ and $k > 2$ one can select global normalized nonnegative finite-entropy weak solutions $(u_\varepsilon, \psi_\varepsilon)$ of (1.2)–(1.4), for $\varepsilon > 0$, and a sequence $\varepsilon_j \downarrow 0$ such that, on every finite interval $(0,T)$,*

$$u_{\varepsilon_j} \longrightarrow u \qquad \text{strongly in } L^2((0,T)\times\mathbb{T}^3), \tag{1.7}$$

$$\psi_{\varepsilon_j} \longrightarrow \psi \qquad \text{strongly in } L^1((0,T)\times\mathbb{T}^3\times B), \tag{1.8}$$

$$\tau(\psi_{\varepsilon_j}) \longrightarrow \tau(\psi) \qquad \text{strongly in } L^1((0,T)\times\mathbb{T}^3). \tag{1.9}$$

The pair (u,ψ) is a global weak solution of the same full system with $\varepsilon = 0$. Its spatial transport, configuration drag, and configuration diffusion fluxes are, respectively, $u\psi$, $(\nabla_x u)q\psi$, and $M\nabla_q(\psi/M)$, and the corresponding positive-diffusion fluxes converge to them in distributions. The zero configuration-flux condition and the prescribed initial data are retained. The limiting pair satisfies (1.6) with $\varepsilon = 0$ and

$$u \in L^\infty_{\mathrm{loc}}([0,\infty);L^2(\mathbb{T}^3)) \cap L^2_{\mathrm{loc}}([0,\infty);H^1(\mathbb{T}^3)), \qquad \psi \in C_{\mathrm{weak}}([0,\infty);L^1(\mathbb{T}^3\times B)).$$

The density has a strong L^1 initial trace. No smallness assumption or strong convergence assumption on the full velocity gradient is imposed.

Only smoothness in the open configuration ball is required of g_0; its derivatives need not extend to ∂B. If a chosen positive-diffusion family has the energy inequality starting at almost every positive time, that additional inequality passes to the limit. We do not include this additional selection property in the unconditional assertion of Theorem 1.1.

2 Estimates before the configuration-drag limit

The fixed-positive-diffusion construction and several configuration estimates will be used below. We spell out their spatial domain and coefficient choices, since existence at $\varepsilon = 0$ alone does not give the limit in this paper.

2.1 Positive diffusion and periodic geometry

Barrett and Süli [1, Theorem 6.1] construct weak solutions at fixed $\varepsilon > 0$ by entropy truncation and time discretization. In their one-spring notation take the scalar potential $V(z) = -k\log(1-2z)$, $0 \le z < 1/2$, so that $V'(z) = 2k/(1-2z)$. With $d(q) = 1 - |q|$, the Maxwellian is comparable to d^k and $k \le dV' \le 2k$. Their potential assumptions therefore hold with exponent $k > 2$. Choose configuration diffusion coefficient one and stress coefficient one. Their isotropic stress subtraction is Id for number density one; its spatial divergence is zero.

The spatial boundary condition in [1] is not periodic. Here is the adaptation of that fixed-ε construction. Replace its velocity space by the closure of mean-zero solenoidal trigonometric polynomials and its density space by $H^1(\mathbb{T}^3\times B, M)$. The velocity projectors preserve solenoidality, periodic integration by parts cancels transport, and mean-zero Poincaré replaces the zero-boundary Poincaré inequality. The spatial compact embeddings follow from periodic charts; the configuration form domain and its compact embedding are unchanged. The finite-cutoff variational existence, positivity, and mass tests use precisely

these properties. Terms that vanished on the spatial boundary now cancel between opposite faces. The configuration tests and the potential inequalities remain the same.

For completeness, use the cutoff entropy with second derivative $1/\min(s, L)$. Its configuration-drag term cancels the full Kramers stress work. At zero body force no forcing estimate consumes viscous dissipation. When $L > \|g_0\|_\infty$, the initial cutoff entropy equals the true initial entropy. The coupled approximation thus retains (1.6) with its displayed constants. At fixed ε, the compactness and cutoff passages in [1] use the same weighted configuration and spatial compact embeddings on these periodic spaces. Strong L^1 density convergence gives strong L^2 convergence of bounded configuration moments by normalization; these moments pair with the weak L^2 velocity gradients. The Hardy estimate below removes stress collars. Lower semicontinuity gives (1.6). Globally extended approximations and nested extraction on integer intervals give a global solution. For a nonzero initial mean c, translate x by ct and add c to the mean-zero velocity. Both equations are preserved, and the constant mean kinetic energy cancels from (1.6). This is a periodic adaptation of a fixed-positive-diffusion construction, not an imported vanishing-diffusion assertion.

2.2 Configuration boundary control and velocity compactness

We use the quadratic Hardy estimate and the weighted Sobolev gain from [2, Lemma 3.2(ii), Proposition 3.7]:

$$\int_B \frac{Mh^2}{d^2} \le C_k \int_B M(|\nabla_q h|^2 + h^2), \tag{2.1}$$

$$\|h\|_{L^{p_*}_M} \le C_k \|h\|_{H^1_M} \quad \text{for some } p_* > 2. \tag{2.2}$$

The first holds for $k > 1$. Fix a finite interval and put $f_\varepsilon = \sqrt{\psi_\varepsilon/M}$ and $i_\varepsilon(t,x) = \int_B M|\nabla_q f_\varepsilon|^2$. Normalization and (2.1) imply

$$\int_B \frac{\psi_\varepsilon}{d^2} \le C(1 + i_\varepsilon), \qquad \int_{d<r} \frac{\psi_\varepsilon}{d} \le Cr(1 + i_\varepsilon). \tag{2.3}$$

Integration by parts, justified by these collars, gives

$$\tau(\psi_\varepsilon) = \mathrm{Id} + 2\int_B Mq \otimes (f_\varepsilon \nabla_q f_\varepsilon), \qquad |\tau(\psi_\varepsilon) - \mathrm{Id}| \le 2\sqrt{i_\varepsilon}. \tag{2.4}$$

In particular the stress is bounded in $L^2_{t,x}$. The equilibrium term Id in this formula is essential.

The energy inequality and momentum equation bound $\partial_t u_\varepsilon$ in $L^{4/3}_t H^{-1}_x$, after solenoidal projection. Indeed the three-dimensional energy interpolation controls $u \otimes u$, and viscosity and stress use their L^2 bounds. Project onto finitely many periodic Laplace modes. Every coefficient has bounded $L^{4/3}$ time derivative and is uniformly Hölder continuous with exponent $1/4$. A finite-dimensional subsequence converges uniformly in time. The integrated squared tail beyond spatial frequency L is at most C/L^2 by the gradient bound and Bessel's inequality. A diagonal in L proves strong $L^2_{t,x}$ compactness. We henceforth write

$$\varepsilon_n \downarrow 0, \quad u_n \to u \text{ in } L^2_{t,x}, \quad A_n := \nabla_x u_n \rightharpoonup A := \nabla_x u \text{ in } L^2_{t,x}, \quad \psi_n \rightharpoonup \psi \text{ in } L^1_{t,x,q}. \tag{2.5}$$

The last extraction follows from entropy uniform integrability on the finite domain. Normalization passes by bounded (t,x) tests. Since $|K| \le C/d$, (2.3) identifies the weak stress by bounded kernel truncation.

The spatial transport already passes: for bounded smooth ϕ,

$$\left|\int (u_n - u)\psi_n \phi\right| \le \|\phi\|_\infty \|u_n - u\|_{L^1_{t,x}}.$$

In the term containing $u(\psi_n - \psi)$, truncate u, use weak L^1 convergence, then remove the truncation using unit configuration mass. Similarly $\varepsilon_n \Delta_x \psi_n$ tends to zero against fixed tests at order ε_n. What remains is the product of A_n with bounded configuration moments of ψ_n. Neither factor is yet strongly compact.

3 Conditioning on the common spatial Brownian motion

The nonlinear quantities used to remove the remaining correlation will be applied to a conditional density. Throughout this section the velocity v is prescribed, deterministic, divergence-free, and belongs to $L^2_t H^1_x$; $A = \nabla_x v$ and $\varepsilon > 0$ are fixed.

3.1 The fibre equation and its measurable flow representation

Let W be one standard three-dimensional Brownian motion, common to every spatial label and every configuration. Put $Z = \sqrt{2\varepsilon} W$ and let Y be the regular Lagrangian flow of $v(t, y + Z(t))$. Then $X = Y + Z$ satisfies

$$\mathrm{d}X = v(t, X)\,\mathrm{d}t + \sqrt{2\varepsilon}\,\mathrm{d}W, \qquad X(0, b) = b. \tag{3.1}$$

The flow preserves volume for almost every path. The deterministic Sobolev flow facts needed here, including stability and the characteristic meaning of bounded transport inequalities with L^1 source, are [2, Appendix A]. Translation preserves all their spatial Sobolev bounds. Smooth approximation gives the adapted version of this flow. The related scalar stochastic representation on the torus is described in [3, Section 3.3]; the configuration fibres below are an additional part of the present construction.

For any trace-free matrix history $\mathsf{B} \in L^2(0, T)$, the weighted variational problem is

$$\langle \partial_t r, h\rangle + \int_B M \nabla_q r \cdot \nabla_q h = \int_B M \mathsf{B}(t) q r \cdot \nabla_q h, \qquad r(0) = r_0 \in L^2_M. \tag{3.2}$$

A dense Galerkin basis of the form domain, including constants, gives

$$\tfrac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|r\|^2_{L^2_M} + \tfrac{1}{2}\|\nabla_q r\|^2_{L^2_M} \le \tfrac{1}{2}|\mathsf{B}|^2 \|r\|^2_{L^2_M}. \tag{3.3}$$

Weak extraction yields $r \in L^\infty_t L^2_M \cap L^2_t H^1_M$ and $\partial_t r \in L^2_t (H^1_M)'$. Testing with the negative part proves positivity; the constant test conserves mass. The same estimate for a difference proves uniqueness and time regularization gives a continuous L^2_M representative. Galerkin measurability and uniqueness give measurable, progressive solutions for measurable progressive histories.

Use $\mathsf{B}(t) = A(t, X(t, b))$ and $r_0 = g_0(b, \cdot)$. Volume preservation and Fubini give $\mathsf{B} \in L^2_t$ on almost every label and path. Define

$$\rho^\omega(t, x, q) = M(q) r^\omega(t, X_t^{-1}(x), q). \tag{3.4}$$

Each density has unit configuration mass. Its labelwise L^2_M bound may grow exponentially with the individual history; we do not claim a global L^2 bound in the labels or paths. The uniform global estimates below use entropy instead. Itô's formula against a smooth test of X, with the fibre equation, shows that $\mathbb{E}\rho$ solves the prescribed-velocity version of (1.4).

3.2 Conditional entropy and logarithmic moments

Applying the configuration entropy calculation to (3.2), then integrating the labels, gives for almost every path

$$\mathrm{Ent}_M(\rho^\omega(t)) + 2\int_0^t I_q(\rho^\omega)\,\mathrm{d}s \le \mathrm{Ent}_M(\psi_0) + C\int_0^t \int_{\mathbb{T}}^3 |A|^2. \tag{3.5}$$

The constant is deterministic: volume preservation makes the last integral the same on every path. In deriving this estimate use (2.4) and absorb $2|A|\sqrt{i_q}$ into Fisher dissipation. No conditional fluid equation is used.

Fix $a = e^4$ and introduce, for each conditional profile,

$$\begin{aligned} s = \rho/M + a, \quad f = \sqrt{s}, \quad G = \Theta(s) = \sqrt{s\log s}, \\ \mathcal{H}_1 = \int_B Ms\log s, \\ \mathcal{H}_2 = \left(\int_B Ms[(\log s)^2 - 2\log s + 2]\right)^{1/2}. \end{aligned} \tag{3.6}$$

The shift keeps all scalar derivatives in a fixed positive range. Along a label $\mathcal{H}_2$ is absolutely continuous with derivative

$$F = -\frac{4}{\mathcal{H}_2}\int_B M\log s|\nabla_q f|^2 + \frac{1}{\mathcal{H}_2}\int_B M(\mathsf{B}:K)\left[s\log s - s - \frac{a}{2}\log^2 s\right]. \tag{3.7}$$

Indeed the derivative of $s[(\log s)^2 - 2\log s + 2]$ is $\log^2 s$ and its second derivative is $2\log s/s$. Integration by parts gives (3.7). The unshifted drag part is alternatively $2\mathcal{H}_2^{-1}\int M\mathsf{B}qf\log s\nabla_q f$. Since $\mathcal{H}_1 \le C\mathcal{H}_2$, it is bounded by $C|\mathsf{B}|(\int M\log s|\nabla f|^2/\mathcal{H}_2)^{1/2}$. The shifted part is bounded by $C|\mathsf{B}|/\mathcal{H}_2$, since

$$\int_B \frac{M\log^2 s}{d} \le \left(\int_B \frac{M}{d^2}\right)^{1/2}\left(\int_B M\log^4 s\right)^{1/2} \le C_{k,a}.$$

Here $\int_B Ms = a+1$, $k > 1$, and $\log^4 s \le C_a s$. The gradient of G is $(\sqrt{\log s} + 1/\sqrt{\log s})\nabla f$, whose square is comparable to $\log s|\nabla f|^2$. Young's inequality, the positive lower bound on $\mathcal{H}_2$, and (3.7) now give

$$\sup_t \int_{\mathbb{T}}^3 \mathcal{H}_2^\omega + \int_0^T\int_{\mathbb{T}}^3 \frac{\int_B M|\nabla_q G^\omega|^2}{\mathcal{H}_2^\omega} + \int_0^T\int_{\mathbb{T}}^3 |F^\omega| \le C. \tag{3.8}$$

All these calculations follow first from convex truncations with derivative $\log^2(\min(s,R))$ and then from monotone convergence and Fatou. Thus they hold in the fibre class without boundary regularity of the initial datum.

Interpolating (2.2) with the fixed L^2_M mass of f gives a full time-space-configuration exponent $r_0 = 4 - 4/p_* > 2$ for f. Every slightly smaller exponent above two also controls G. Moreover $\mathcal{H}_2^2 \le C\int_B Mf^{2+\zeta}$ for sufficiently small $\zeta > 0$, so $\mathcal{H}_2$ is bounded in $L^2_{t,x}$, uniformly in the path. The moments $\mathcal{H}_1, \mathcal{H}_2$ and G^2 are uniformly integrable on the joint probability-time-space domain. These are integrability estimates, not spatial compactness assertions.

3.3 Identification with the coupled density

We next justify that this conditional construction represents the actual positive-diffusion coupled solution, rather than another solution of its prescribed-velocity equation.

Lemma 3.1. *Fix $\varepsilon > 0$, $k > 1$, and a divergence-free $v \in L^2_t H^1_x$. Two normalized nonnegative solutions of its linear Fokker–Planck equation coincide if both have finite configuration and ε-weighted spatial Fisher information and the same strong L^1 initial trace.*

Proof. For densities p_i, put $b = (v, Aq - \nabla U)$ and $D_\varepsilon = \operatorname{diag}(\varepsilon \mathrm{Id}, \mathrm{Id})$. Hardy and normalization give

$$\sqrt{p_i} \in L^2_t H^1_{x,q}, \qquad \int |b|^2 p_i \le C \left(\int |v|^2 + \int |A|^2 + \int p_i / d^2 \right) < \infty.$$

For $\delta > 0$, set $s_i = p_i + \delta M$ and

$$J(s_1, s_2) = \frac{(s_1 - s_2)^2}{s_1 + s_2}, \qquad z = \frac{s_1 - s_2}{s_1 + s_2}.$$

The function J is nonnegative, convex, one-homogeneous and at most $s_1 + s_2$. Its derivatives are $J_1 = 2z - z^2$, $J_2 = -2z - z^2$, hence $|J_i| \le 3$, and

$$D^2 J = \frac{2 w \otimes w}{s_1 + s_2}, \qquad w = (1 - z, -1 - z), \qquad \sum_i s_i |\partial_\alpha J_i|^2 \le 2\, D^2 J[\partial_\alpha s, \partial_\alpha s]. \tag{3.9}$$

The common conservative drift cancels by homogeneity. Both shifted equations have source $\delta \operatorname{div}_q(AqM) = -\delta M A : K$, whose contribution is at most $C\delta \int |A|$.

Here are the required weak chain-rule details. First cut off configurations at distance r from the boundary and mollify in (x, q) at radius $h < r/4$. Rewriting the mollified equation with the original b creates flux error

$$R_{i,h}(\xi) = \int [b(y) - b(\xi)] s_i(y) \eta_h(\xi - y)\, \mathrm{d}y, \qquad \frac{|R_{i,h}|^2}{s_{i,h}} \le \int |b(y) - b(\xi)|^2 s_i(y) \eta_h(\xi - y)\, \mathrm{d}y.$$

Its integral on the fixed interior tends to zero: the v term uses L^2 translation continuity and configuration mass $1 + \delta$; for Aq split $(A(y_x) - A(x)) y_q + A(x)(y_q - q)$ and use the same continuity and an $O(h^2 \int |A|^2)$ term. The field ∇U is Lipschitz on that interior. No spatial derivative of A is taken. By (3.9) the entropy error is bounded by half the diffusion dissipation plus $C/\min(\varepsilon, 1) \sum_i |R_{i,h}|^2 / s_{i,h}$. The divergence error against the fixed cutoff vanishes as well. Let $h \to 0$ at fixed r, δ, ε; retain the nonnegative dissipation by lower semicontinuity. Bounded J_i, the L^1 diffusion fluxes, and the bound on $\int |b|^2 p_i$ justify the remaining passages by drift truncation and time regularization.

Choose the configuration cutoff with derivative at most C/d on $d < 2r$. The diffusion collar is bounded by

$$C \sum_i \left(\int_{d<2r} s_i / d^2 \right)^{1/2} \left(\int_{d<2r} |\nabla_q \sqrt{s_i}|^2 \right)^{1/2} \to 0.$$

The confining collar is bounded by $C \int_{d<2r} \sum_i s_i / d^2 \to 0$; the matrix-drift collar by $C \sum_i (\int |A|^2 s_i)^{1/2} (\int_{d<2r} s_i / d^2)^{1/2} \to 0$. All these integrals include time and space. Thus the common initial trace gives

$$\int J(p_1 + \delta M, p_2 + \delta M)(t) \le C\delta \int_0^t \int_{\mathbb{T}^3} |A|.$$

Dominated convergence as $\delta \to 0$ yields $p_1 = p_2$. □

To obtain the Fisher assumptions for $\mathbb{E}\rho$, approximate v strongly in $L^2_t H^1_x$ by smooth divergence-free v_m. For fixed m, A_m is bounded; (3.3), volume preservation and Jensen put the averaged $g_m = \mathbb{E}\rho_m/M$ in $L^\infty_t L^2_{x,q,M}$ with its configuration form bound. Mollify its weak equation only in x. The v_m and A_m commutators tend to zero in L^2, since their spatial derivatives are bounded and g_m is L^2. Testing by the spatially mollified g_m gives the ε spatial form bound. Removing this mollification and then the convex entropy regularization gives

$$\mathrm{Ent}_M(\mathbb{E}\rho_m(t)) + 2\int_0^t I_q(\mathbb{E}\rho_m) + 4\varepsilon\int_0^t I_x(\mathbb{E}\rho_m) \le \mathrm{Ent}_M(\psi_0) + C\int_0^t\int_{\mathbb{T}}^3 |A_m|^2. \tag{3.10}$$

Its constant is independent of m; no uniqueness assertion has yet been used. The original bounded initial density is retained by the fibre L^2 energy estimate when removing smooth initial approximations.

The flows X_m converge in measure. Their pulled-back gradients converge strongly in L^2, since

$$A_m(X_m) - A(X) = (A_m - A)(X_m) + A(X_m) - A(X);$$

volume preservation controls the first term and bounded continuous approximation of A controls the second. The homogeneous entropy comparison above, now with two matrix histories, absorbs their difference into the configuration dissipation and gives on each fibre

$$\int_B J(Mr_m, Mr_\ell)(t)\,\mathrm{d}q \le C\int_0^t |A_m(X_m) - A_\ell(X_\ell)|^2\,\mathrm{d}s.$$

Its label integral and Cauchy–Schwarz imply strong L^1 convergence of the fibre densities. Returning by the converging inverse flows identifies their Eulerian limit. Lower semicontinuity in (3.10) gives both Fisher bounds for $\mathbb{E}\rho$. These also give its strong initial trace, either by entropy or by comparison with the zero-matrix configuration semigroup.

The coupled solution has the same trace. In fact $\log g_0$ is bounded, weak continuity and (1.6) give $\limsup_{t\downarrow 0} \mathrm{Ent}_M(\psi(t)) \le \mathrm{Ent}_M(\psi_0)$, and hence

$$\mathrm{Ent}(\psi(t) \mid \psi_0) = \mathrm{Ent}_M(\psi(t)) - \int \psi(t)\log g_0 \longrightarrow 0.$$

Pinsker's inequality gives strong L^1 convergence. Lemma 3.1 now proves $\psi_\varepsilon = \mathbb{E}\rho^\omega$. In particular the conditioning has not changed the deterministic stress feedback in the fluid equation.

3.4 Nonlinear transforms before expectation

If a fibre functional $B(t,b)$ has finite variation with source S_B, test its Eulerian push-forward against $\phi(t, X(t,b))$. Itô's product formula has no quadratic covariation with B. Its sole correction is $\varepsilon\int B\Delta_x\phi(t,X)\,\mathrm{d}b$; its martingale coefficient is $\sqrt{2\varepsilon}\int B\nabla_x\phi(t,X)\,\mathrm{d}b$. For bounded functionals this coefficient is bounded. For $\mathcal{H}_1, \mathcal{H}_2$ their deterministic global pathwise mass bounds make it square-integrable. Fubini and expectation therefore give

$$(\partial_t + v\cdot\nabla_x)\mathbb{E}B = \varepsilon\Delta_x\mathbb{E}B + \mathbb{E}S_B. \tag{3.11}$$

For G first keep a fixed smooth configuration test and use the fibre variational equation. Smooth fibre and spatial-flow approximations justify the same identity; their fluxes and sources pass by the preceding estimates. There is no spatial derivative of the conditional density in this argument.

For $\vartheta_\kappa(z) = z/(1+\kappa z)$ apply (3.11) to

$$B_{\delta,\kappa} = \frac{\vartheta_\kappa(\mathcal{H}_1)}{1+\delta\mathcal{H}_2}, \qquad W_{\delta,\kappa} = (1+\delta\mathcal{H}_2)(1+\kappa\mathcal{H}_1)^2. \tag{3.12}$$

The stress-work term is $A : \mathbb{E}[\tau(Ms)/W_{\delta,\kappa}]$, and the configuration Fisher term has precisely the same conditional weight. The denominator derivative is bounded in L^1 by $C\delta/\kappa$, using (3.8). Thus δ will be removed before κ. The bound $B_{\delta,\kappa} \le 1/\kappa$ ensures that its spatial Laplacian vanishes against fixed tests when $\varepsilon \to 0$. This identity concerns $\mathbb{E}B(\rho)$, not $B(\mathbb{E}\rho)$. Applying the latter nonlinear functional directly to the spatially diffusive density would introduce spatial Hessian terms with no required sign.

4 A transported envelope for the limiting conditional moments

Return to the coupled family in (2.5) and its conditional representation. Along each random characteristic define

$$Q_n^\omega(t, X_n(t,b)) = \mathcal{H}_{2,0}(b) + \int_0^t |F_n^\omega(s, X_n(s,b))|\,\mathrm{d}s. \tag{4.1}$$

It dominates $\mathcal{H}_{2,n}^\omega$, and its spatial mass is bounded by the same deterministic constant on every path. For $\beta_L(z) = Lz/(L+z)$, $q_{n,L} = \mathbb{E}\beta_L(Q_n^\omega)$ satisfies

$$(\partial_t + u_n\cdot\nabla_x)q_{n,L} = \varepsilon_n\Delta_x q_{n,L} + \mathbb{E}[\beta_L'(Q_n^\omega)|F_n^\omega|] \ge \varepsilon_n\Delta_x q_{n,L}. \tag{4.2}$$

At fixed L it is bounded, so strong velocity convergence passes its transport and the small Laplacian vanishes. The limit q_L is nondecreasing along the regular Lagrangian flow of u. Its increasing supremum Q is finite almost everywhere by its common mass bound. Put $R = \operatorname{wlim}\mathbb{E}\mathcal{H}_{2,n}^\omega$. The L^2 bound on $\mathcal{H}_{2,n}^\omega$ and $\mathcal{H}_2 - \beta_L(\mathcal{H}_2) \le \mathcal{H}_2^2/L$ show that $R \le Q$.

On $[0,T_0]$ choose a later time $T_1 > T_0$ at which $Q(T_1)$ has finite mass, and transport $1+Q(T_1)$ backward by the limiting flow. This defines

$$N \ge 1, \qquad R \le N, \qquad N(t, X(t,b)) = N(0,b) < \infty \quad \text{a.e.} \tag{4.3}$$

For this possibly unbounded N, transport means characteristic constancy and transport of its bounded Borel functions. No raw product uN is used. The normalizer is introduced after the zero-diffusion limit; its later-time definition is never an anticipating stochastic test.

Let $y = \operatorname{wlim}\mathbb{E}G_n^\omega$. Weighted Cauchy–Schwarz yields

$$\frac{\int_B M|\nabla_q\mathbb{E}G_n^\omega|^2}{\mathbb{E}\mathcal{H}_{2,n}^\omega} \le \mathbb{E}\frac{\int_B M|\nabla_q G_n^\omega|^2}{\mathcal{H}_{2,n}^\omega}.$$

Convexity of $z \mapsto z\sqrt{\log z^2}$ on the shifted range and the L^2 triangle inequality also give $\|\mathbb{E}G_n^\omega\sqrt{\log(\mathbb{E}G_n^\omega)^2}\|_{L_M^2} \le C\mathbb{E}\mathcal{H}_{2,n}^\omega$. Convex lower semicontinuity, (3.8), and $R \le N$ imply

$$\int_0^{T_0}\int_{\mathbb{T}}^3 \frac{\mathcal{D}_y}{N} \le C, \qquad \mathcal{D}_y = \int_B M|\nabla_q y|^2, \qquad h := y/N^2 \in L_{t,x}^\infty L_M^2 \cap L_{t,x}^2 H_M^1. \tag{4.4}$$

These are the energy bounds needed later for deterministic configuration renormalization. Only bounded envelope truncations have entered a spatial transport limit.

5 Conditional cutoffs and the stress-work defect

The stress-work identity must use the same conditional cutoffs as the entropy equation. For $c = (\delta, \kappa)$, define

$$T_{n,c} = \mathbb{E}\frac{\tau(Ms_n^\omega)}{W_{n,c}^\omega}, \qquad z_{n,c} = A_n \mathbb{E}\frac{s_n^\omega}{W_{n,c}^\omega}, \qquad W_{n,c}^\omega = (1+\delta\mathcal{H}_{2,n}^\omega)(1+\kappa\mathcal{H}_{1,n}^\omega)^2. \tag{5.1}$$

The raw shifted stress is $\tau_n^a = \tau(\psi_n) + a\mathrm{Id}$. All cutoff removals below first take $n \to \infty$ at fixed c, then $\delta \downarrow 0$, and then $\kappa \downarrow 0$.

5.1 Integrable stress squares and a common drag direction

For a conditional profile let $S = \int_B Ms/d$, $J_q = \int_B Ms\log s/d$ and $D_G = \int_B M|\nabla_q G|^2$. Applying (2.1) to G gives

$$J_q^2 \le C\mathcal{H}_1(D_G + \mathcal{H}_1).$$

Jensen's inequality for the finite measure $M\,\mathrm{d}q/d$ gives $S^2\log^2(2+S) \le C(1+J_q^2)$. Since $\mathcal{H}_1 \le C\mathcal{H}_2$,

$$\left|\frac{\tau(Ms)}{W_c}\right|^2 \log^2\left(2+\left|\frac{\tau(Ms)}{W_c}\right|\right) \le C_\delta(1 + D_G/\mathcal{H}_2), \tag{5.2}$$

$$\int_B \frac{M}{d}\,\Phi(|A|s/W_c) \le C_\delta(1+|A|^2 + D_G/\mathcal{H}_2), \qquad \Phi(z) = z\log(1+z). \tag{5.3}$$

For the second inequality one also uses $bc\log(2+b) \le C[b^2 + c^2\log^2(2+c) + 1]$ for $b, c \ge 0$. The right sides have uniformly bounded integrals by (3.8). The superlinear factors on the left therefore give uniform integrability. Jensen transfers it to $|T_{n,c}|^2$ in (t,x) and to $z_{n,c}$ in $M\,\mathrm{d}q\,\mathrm{d}x\,\mathrm{d}t/d$.

The conditional raw stress is bounded in joint L^2. The cutoff factors approach one in measure uniformly in n by the L^1 moment bounds. Consequently

$$\sup_n \|T_{n,c} - \tau_n^a\|_{L^p_{t,x}} \longrightarrow 0 \qquad (1 < p < 2). \tag{5.4}$$

The radial matrix K is positive semidefinite. The $T_{n,c}$ increase in the Loewner order when their scalar cutoffs are removed, and their squared Frobenius norms increase as well, since $|S+D|^2 - |S|^2 = 2\,\mathrm{tr}(SD) + |D|^2 \ge 0$ for positive semidefinite S, D. At fixed (t,x,q), all $z_{n,c}$ have the common matrix direction A_n and nonnegative ordered scalar magnitudes. Thus for two ordered cutoffs

$$|z_{n,c} - z_{n,c'}| = |z_{n,c}| - |z_{n,c'}| \quad \text{when } W_{n,c} \le W_{n,c'}. \tag{5.5}$$

Expectation has preserved both monotonicities needed here.

For clarity, this identifies a weighted limit of the raw drag, not merely a cutoff-dependent product. The raw $A_n s_n$ are uniformly integrable in the unweighted M measure: the superlinear function $\Phi_{1/2}(z) = z\sqrt{\log(1+z)}$ obeys

$$\Phi_{1/2}(|A|s) \le Cs(1 + |A|^2 + \log s).$$

Normalization makes its right side integrable after averaging. Also, by (2.1),

$$\int M|A_n|\mathbb{E}s_n/d \le C\int(1 + |A_n|^2 + \mathbb{E}I_q(\rho_n)).$$

At fixed cutoffs extract weighted weak L^1 limits z_c and g_c of $z_{n,c}$ and $|z_{n,c}|$. Equation (5.5) implies $|z_c - z_{c'}| \le g_c - g_{c'}$. The increasing g_c have bounded weighted mass, so z_c are Cauchy in weighted L^1. Their limit coincides with the raw weak product by the unweighted uniform integrability. Denote

$$\beta = \operatorname{wlim}(A_n\psi_n) - A\psi. \tag{5.6}$$

The work covariance is therefore exactly $\int_B K : \beta$, with no second product selected by removing the cutoffs.

5.2 Periodic linear estimates and the compact convective part

Split $u_n = v_n + w_n$. Here v_n has zero initial data and solves the periodic solenoidal Stokes equation with force $\operatorname{div}_x \tau(\psi_n)$; w_n has initial data u_0 and force $-u_n \cdot \nabla_x u_n$. Let $v_{n,c}$ be the analogous zero-initial Stokes solution forced by $\operatorname{div}_x T_{n,c}$. We need the linear L^p gradient estimate for stress forcing and the second-derivative estimate for vector forcing, $1 < p < 2$, used in [2, Section 5.4]. We record why their spatially periodic versions hold.

The causal kernels in both estimates have the form $D_x^2 \mathbb{P} G_\nu$, where G_ν is the heat kernel and $\mathbb{P}$ is the solenoidal projector. They are sums of second Gaussian derivatives and fourth derivatives of the Newton-potential convolution with a Gaussian. For $0 < t < T$ and sufficiently large $|x|$ their absolute values are at most $C_{\nu,T}|x|^{-5}$. To see this, differentiate the Newton kernel four times and split its convolution at $|y| = |x|/2$; the remaining Gaussian tail has faster decay. Thus the space-time tails of these kernels are integrable and tend to zero outside large spatial balls.

Cut a smooth periodic forcing, extended to $\mathbb{R}^3$, off to a cube Q_L. On its interior at distance $\sqrt{L}$ from the boundary, the whole-space and periodic convolutions differ by at most the forcing supremum times the kernel tail. Apply the whole-space L^p estimate, divide its pth-power bound by the number of periodic cells, and let $L \to \infty$. The interior fraction tends to one and the error tends to zero. Density extends the resulting periodic bound to all L^p forcings. This transfers a linear estimate only; it does not transfer a nonlinear existence theorem between domains.

The convective force belongs to $L^{5/4}_{t,x}$ by the fluid energy bounds. The transferred estimate gives $w_n \in L^{5/4}_t W^{2,5/4}_x$. Its mean-zero pressure is $-(-\Delta)^{-1}\operatorname{div}(-u_n \cdot \nabla u_n)$; the periodic $|x|^{-2}$ kernel is integrable, giving an $L^{5/4}$ bound. Hence $\partial_t w_n$ is bounded in $L^{5/4}_t W^{-1,5/4}_x$. Spatial smoothing and finite-dimensional time compactness give compactness of ∇w_n in $L^{5/4}_{t,x}$; interpolation with its L^2 bound gives

$$\nabla w_n \longrightarrow \nabla w \quad \text{in } L^p_{t,x}, \qquad p < 2. \tag{5.7}$$

The fixed initial-data Stokes part is common to all n. From (5.4) and the stress-to-gradient estimate, $\nabla v_{n,c} - \nabla v_n \to 0$ in L^p, uniformly in n, as c is removed.

5.3 Local Stokes work and retained oscillation

At fixed c, write $T_c = \operatorname{wlim} T_{n,c}$ and $G_c = \operatorname{wlim} |T_{n,c}|^2$. By (5.2), G_c is L^1. The G_c increase and have bounded masses, so $G_c \uparrow G$ in L^1 for a finite G. The needed tail comparison concerns these weak limits. It does not assert uniform integrability of the raw stress squares. Indeed use the monotone caps $\min(|T_{n,c}|^2, L^2)$. Their weak limits increase both with L and with cutoff removal; the two suprema commute and equal G. Given $\zeta > 0$, choose c_0, L_0 so that the gap from G to the doubly capped weak limit has L^1 norm less than ζ. For every smaller c, the weak limiting energy in $|T_{n,c}| > 2L_0$ is at most $4\zeta/3$, since

$$b^2 \mathbf{1}_{b>2L} \le \tfrac{4}{3}\{b^2 - \min(b^2, L^2)\}.$$

Pair a bounded capped stress with the L^p-small difference of the cut and uncut Stokes gradients, then remove the cap with this tail comparison and the uniform L^2 gradient bounds. It follows that the weak work limits $T_{n,c} : \nabla v_{n,c}$ and $T_{n,c} : \nabla v_n$ have the same L^1 limit as c is removed. Equation (5.7) and fixed-c uniform integrability also give $\operatorname{wlim}(T_{n,c} : \nabla w_n) = T_c : \nabla w$.

On the other hand $0 \le T_c \le \tau(\psi) + a\mathrm{Id}$ in the Loewner order, and (5.4) gives $T_c \to \tau(\psi) + a\mathrm{Id}$ in L^p, $p < 2$. Domination by this L^2 stress gives strong L^2 convergence, hence strong L^2 convergence of the limiting cutoff Stokes gradients.

For fixed c, test the Stokes equation by its velocity times a smooth space-time cutoff. Its energy-class velocity is strongly continuous in L^2; pressure and force fluxes pass by strong velocity and weak pressure/stress convergence. Comparing the local energy identities gives

$$\operatorname{wlim}(T_{n,c} : \nabla v_{n,c}) = T_c : \nabla v_c - \nu\mu_c, \qquad \mu_c = \operatorname{wlim} |\nabla v_{n,c}|^2 - |\nabla v_c|^2 \ge 0. \tag{5.8}$$

Remove the cutoffs by the preceding capped-work comparison and the common-direction argument of (5.5). The total work limit is $A : (\tau(\psi) + a\mathrm{Id}) + \int_B K : \beta$ by (5.6). Subtract the compact w work from (5.8). We obtain

$$\nu\mu = -\int_B K : \beta, \qquad \mu \ge 0, \qquad \mu \in L^1_{t,x}. \tag{5.9}$$

The last assertion follows because the resulting work terms are L^1 functions. The isotropic shift contracts to zero with the trace-free A. This is the local-energy comparison of [2, Proposition 5.4], with the conditional PSD, cap, and common-direction properties verified above.

We will also use its oscillation consequence. If Π is the velocity-gradient marginal of the joint law constructed below, then

$$\langle |\Pi - A|^2 \rangle \le \mu. \tag{5.10}$$

Here is a bounded-moment way to check this consequence. For a centered gradient matrix use $P_R(H) = \min(|H|^2, R)$, whose Lipschitz constant is at most $2\sqrt{R}$. The L^p-small cut-gradient difference makes their P_R moments coincide in the cutoff limit. The cut-gradient square defect bounds its centered P_R moment. Remove the cutoff first, then let $R \to \infty$. The compact ∇w_n adds no oscillation. This proves (5.10); it does not identify μ with the entire viscous dissipation defect of u_n.

6 Joint conditional fluctuations and configuration coercivity

We now compare the weak limit of the conditional entropy with the square of its weakly limiting square root. Define

$$f = \operatorname{wlim} \mathbb{E} f_n^\omega, \quad y = \operatorname{wlim} \mathbb{E} G_n^\omega, \quad \theta = \operatorname{wlim} \mathbb{E} \mathcal{H}_{1,n}^\omega, \quad \eta = \theta - \int_B M y^2 \ge 0. \tag{6.1}$$

In particular $\eta \le CN$ by (4.3) and $\mathcal{H}_1 \le C\mathcal{H}_2$. To retain correlations with A_n, construct at each (t, x, q) the joint law of $(A_n, f_n^\omega, \nabla_q f_n^\omega)$ before averaging the conditional variable. One can do this without a Young-measure theorem on Brownian path space: extract weak-star limits of $\mathbb{E}\Phi(A_n, f_n^\omega, \nabla_q f_n^\omega)$ for a countable dense set of compactly supported continuous state tests. Positivity gives a positive functional on the finite-dimensional state space. Riesz representation gives a measure; the moment bounds give tightness and total mass one. Denote its state by (Π, λ, Λ) and its expectation by brackets. The Π marginal is independent of q, is trace-free, has mean A, and satisfies (5.10).

The higher moment of f_n and the L^2 gradient bound make $\lambda\Lambda$ uniformly integrable. They likewise absorb the logarithmic multiplier in the gradient of G. Consequently, with $b = \sqrt{\log \lambda^2}$,

$$f = \langle \lambda \rangle, \qquad \nabla_q f = \langle \Lambda \rangle, \qquad y = \langle \lambda b \rangle, \qquad \nabla_q y = \langle (b + b^{-1})\Lambda \rangle, \tag{6.2}$$
$$\overline{s} := \langle \lambda^2 \rangle = \psi/M + a.$$

The square-root variance $V_f = \langle \lambda^2 \rangle - f^2$ obeys

$$\nabla_q V_f = 2\langle (\lambda - f)(\Lambda - \nabla_q f) \rangle, \qquad |\nabla_q \sqrt{V_f}|^2 \le \langle |\Lambda - \nabla_q f|^2 \rangle. \tag{6.3}$$

Since $\lambda \mapsto \lambda\sqrt{\log \lambda^2}$ has derivative bounded below on $\lambda \ge \sqrt{a}$, the two-copy variance formula gives $\int_B M V_f \le C\eta$.

6.1 Retaining positive Fisher moments

At each fixed conditional entropy cutoff, the positive measures $\mathbb{E}[|\nabla_q f_n^\omega|^2 / W_{n,c}^\omega]$ have weak limits. They increase when the cutoffs are removed and have bounded total masses. Let m be the increasing limit of their absolutely continuous densities. For a bounded state moment $\min(|\Lambda|^2, L^2)$, the cutoff loss is at most

$$L^2 \mathbb{E}(1 - 1/W_{n,c}) \le L^2 \mathbb{E} \min(\delta \mathcal{H}_{2,n} + 2\kappa \mathcal{H}_{1,n}, 1),$$

whose total integral tends to zero as c is removed. Weak passage of this bounded moment, then removal of c and $L \to \infty$, gives

$$m \ge \langle |\Lambda|^2 \rangle \quad \text{a.e.}, \qquad \varpi := \int_B M(m - |\nabla_q f|^2) \ge 0. \tag{6.4}$$

This is a density inequality: for an increasing sequence of finite positive measures, the absolutely continuous part of the limit is the increasing limit of their absolutely continuous parts, since all their singular parts are supported on a countable union of null sets.

The positive transformed source is $-\Theta''(s_n)|\nabla_q s_n|^2$. Its coefficient in the f_n variables is $(b + b^{-3})/\lambda$, bounded and continuous on $\lambda \ge \sqrt{a}$. The same bounded-state argument implies that its absolutely continuous limit V satisfies

$$V \ge \langle (b + b^{-3})|\Lambda|^2/\lambda \rangle. \tag{6.5}$$

The singular positive part remains favorable in a supersolution inequality; we will discard it before using a rough spatial normalizer. From (6.3), (6.4), and Hardy,

$$\int_B \frac{M V_f}{d} \le C\sqrt{\eta}\sqrt{\varpi + \eta}. \tag{6.6}$$

6.2 An all-state two-copy inequality

The following algebra ensures that averaging has not destroyed configuration coercivity. Take two independent states (λ, Λ), (λ', Λ') of the same law, and put $b = \sqrt{\log \lambda^2}$, $c = \sqrt{\log(\lambda')^2}$, $r = \lambda'/\lambda$. Set

$$B_1 = 2 + rc(b + b^{-3}), \quad B_2 = 2 + r^{-1}b(c + c^{-3}), \quad B_3 = (b + b^{-1})(c + c^{-1}).$$

The matrix with diagonal $B_1 - 1, B_2 - 1$ and off-diagonal $-(B_3 - 1)$ is positive semidefinite. Its diagonal entries are positive, and direct expansion of its determinant gives

$$bc[r + r^{-1} + 2 - 2b/c - 2c/b] + 2[b/c + c/b - 2] \\ + (bc)^{-1}[rc^2/b^2 + r^{-1}b^2/c^2 + 2 - 2b/c - 2c/b].$$

By symmetry take $r \geq 1$, $h = \log r$, and $z = \frac{1}{2}\log(c^2/b^2)$. Since $b^2 \geq 4$ and $c^2 - b^2 = 2h$, $0 \leq z \leq h/2$. The first bracket is $2\cosh h + 2 - 4\cosh z \geq 0$, the middle bracket is nonnegative, and the last is $2\cosh(h + 2z) + 2 - 4\cosh z \geq 0$. It follows, for every pair of admissible states, that

$$B_1|\Lambda|^2 + B_2|\Lambda'|^2 - 2B_3\Lambda \cdot \Lambda' \geq |\Lambda - \Lambda'|^2. \tag{6.7}$$

Take the expectation over the two independent copies. The mixed terms factor, using $y = \langle \lambda b \rangle$ and $\nabla_q y = \langle (b + b^{-1})\Lambda \rangle$, to give the pointwise inequality

$$4\langle |\Lambda|^2 \rangle + 2y\langle (b + b^{-3})|\Lambda|^2/\lambda \rangle - 2|\nabla_q y|^2 \geq 2(\langle |\Lambda|^2 \rangle - |\nabla_q f|^2). \tag{6.8}$$

Replace the two positive moments on the left by m and V using (6.4)–(6.5). After subtracting $2(m - |\nabla_q f|^2)$ from the new left side, the retained Fisher excess leaves at least $2(m - \langle |\Lambda|^2 \rangle) \geq 0$. Integration against M gives

$$4\int_B Mm + 2\int_B MyV - 2\mathcal{D}_y \geq 2\varpi. \tag{6.9}$$

The joint-law gradient variance contributes at least twice its mass. The excess $m - \langle |\Lambda|^2 \rangle$ contributes four times its mass, so retained concentration has the favorable sign without being assumed zero.

6.3 Covariances and configuration boundary integrability

Use $s = \lambda^2$ within law brackets and put

$$L = \langle s\Theta'(s) \rangle, \quad P = \langle \Theta'(s) \rangle, \quad \Gamma = \langle \Pi G \rangle - Ay, \quad \Gamma' = \langle \Pi s\Theta'(s) \rangle - AL. \tag{6.10}$$

Cauchy–Schwarz and (5.10) give

$$|\Gamma|^2 \leq \mu \operatorname{Var}(G), \qquad |\Gamma' - \Gamma|^2 \leq \tfrac{1}{4}\mu \operatorname{Var}(G). \tag{6.11}$$

For the second bound note that $s\Theta'(s) - G$, as a function of G, has derivative of absolute value at most $1/2$. Since $\int_B M \operatorname{Var}(G) = \eta \leq CN$, these imply, before any renormalization,

$$\int_B M|\Gamma/N^2|^2 + \int_B M|\Gamma'/N^2|^2 \leq C\mu/N^3. \tag{6.12}$$

Thus the normalized covariance fluxes belong to $L^2(M)$ and, by Cauchy–Schwarz with M/d^2, to $L^1(M/d)$. The mean-gradient flux Aqh is $L^2(M)$ by the uniform q norm in (4.4).

For signed scalar sources a direct profile estimate uses exactly $k > 2$. Choose $2/3 < \alpha < k/(k+1)$. Then $G^{4/3} \leq C_{a,\alpha} s^\alpha$ and

$$\int_B \frac{M}{d} G^{4/3} \leq C\left(\int_B Ms\right)^\alpha \left(\int_B Md^{-1/(1-\alpha)}\right)^{1-\alpha} \leq C_{k,a}. \tag{6.13}$$

Hence $A_n G_n$, $A_n s_n \Theta'(s_n)$, and the lower shifted composites are bounded in the mixed space

$$L^2_{t,x}(L^{4/3}(B, M\,\mathrm{d}q/d)),$$

including the conditional variable. Their signed limits have no concentration in that measure. For the conditional diffusion collar, Cauchy–Schwarz gives

$$\int_{t,x,d<r} \frac{M}{d}|\nabla_q G_n| \le \left(\int \frac{M|\nabla_q G_n|^2}{\mathcal{H}_{2,n}}\right)^{1/2} \left(\int_{t,x} \mathcal{H}_{2,n} \int_{d<r} \frac{M}{d^2}\right)^{1/2} \to 0.$$

The transformed-drag collar is at most $C\|A_n\|_{L^1_{t,x}}(\int_{d<r} M/d)^{1/4} \to 0$ by (6.13). Both vanish uniformly. The normalized limiting fluxes have the same boundary property by (4.4) and (6.12). Consequently the transformed variational inequalities hold for tests meeting the closed configuration ball, not only for interior tests.

7 The normalized entropy-defect inequality

We next justify the limiting transport and configuration operations before estimating their four resulting terms. This order matters because the raw entropy moment θ need not have spatial integrability above exponent two.

7.1 The lower transformed inequality

The averaged G equation has just the linear spatial diffusion term in (3.11). At the limit its positive configuration source is a measure. Discard its singular positive part, retaining the density V of (6.5). Write $D_t = \partial_t + u \cdot \nabla_x$ and $\mathcal{L}_M h = M^{-1}\operatorname{div}_q(M\nabla_q h)$. The resulting inequality is

$$D_t y \ge \mathcal{L}_M y - \frac{1}{M}\operatorname{div}_q(M\langle \Pi G\rangle q) + K : \langle \Pi[(s-a)\Theta'(s) - G]\rangle + V. \tag{7.1}$$

Here $s = \lambda^2$, $G = \Theta(s) = \lambda\sqrt{\log \lambda^2}$ inside the brackets, $\langle \Pi G\rangle = Ay + \Gamma$, and $\langle \Pi s\Theta'(s)\rangle = AL + \Gamma'$. For a single fibre with its trace-free matrix history $\mathsf{B}(t)$, the chain rule giving these signs is

$$\partial_t G = \mathcal{L}_M G - \frac{1}{M}\operatorname{div}_q(M\mathsf{B}qG) + (\mathsf{B} : K)[(s-a)\Theta'(s) - G] - \Theta''(s)|\nabla_q s|^2.$$

Use (3.11) before taking the joint-law limit; the last source is positive and has retained density V. The configuration boundary inequality and lower initial trace $y(0) \ge \Theta(g_0 + a)$ are inherited in the variational sense. Thus (7.1) does not impose a pointwise boundary value or assert equality of a residual measure. Its scalar pairings with fixed nonnegative smooth configuration tests have $L^r_{t,x}$ regularity for some $r > 2$. To interpret these along the limiting spatial flow, mollify in x. The Sobolev transport commutator tends to zero in L^1, since $\nabla u \in L^2$ and the scalar pairing has exponent above two. Apply an increasing bounded scalar truncation, use the bounded mild transport inequality of [2, Proposition A.5], then remove the truncation by dominated convergence for the signed L^1 source and monotone convergence for the positive residual. A countable dense family of nonnegative configuration tests and the flux integrability give a common full-measure set of labels on which the whole configuration inequality holds.

Along each such label N is a fixed finite scalar. Division gives

$$\begin{aligned} D_t h \ge \mathcal{L}_M h - \frac{1}{M}\operatorname{div}_q(M(Ah + \Gamma/N^2)q) \\ + \frac{1}{N^2}K : \langle \Pi[(s-a)\Theta'(s) - G]\rangle + \frac{V}{N^2}. \end{aligned} \tag{7.2}$$

This is the labelwise variational inequality for $h = y/N^2$. By (4.4), (6.12), and (6.13), $h = y/N^2$ belongs to $L^\infty_t L^2_M \cap L^2_t H^1_M$, its force flux is both $L^2(M)$ and $L^1(M/d)$, and its signed scalar source and V/N^2 are $L^1(M)$. The boundary inequality and initial trace are inherited from the conditional variational identities. These are the hypotheses of [2, Proposition B.6], now for a deterministic configuration supersolution on a label. In its ordinary-divergence notation, rewrite the weighted force divergence as ordinary divergence plus the ∇U term; the $L^1(M/d)$ flux bound controls that additional signed source. We use the convex supersolution statement of B.6 here, with its lower initial and boundary inequalities, and do not invoke an equality-class continuity assertion.

Apply smooth convex approximations to the square with bounded nonnegative derivative. Their flux cross products are integrable by the normalized L^2 bounds. For the signed sources note that $L \le Cy$ and P is bounded. Quadratic Hardy and (6.11) bound their products with y/N^4 by

$$C\left[\frac{1+|A|^2}{N^3} + \frac{\mathcal{D}_y}{N^4} + \frac{\mu}{N^3}\right]. \tag{7.3}$$

These are integrable by (4.4). Remove the derivative cap by dominated convergence for signed terms and monotone convergence for yV. No fourth configuration moment of h has been used. Because the singular positive measure was discarded first, no undefined representative of N^{-2} has been multiplied into such a measure.

7.2 The upper entropy inequality

Keep the bounded quotients (3.12) until after normalization. At each fixed δ, κ, their limiting equation gives an upper transport inequality for a bounded scalar, with L^1 signed work and a nonpositive Fisher residual. Discard only the singular negative part of that residual, apply the bounded mild transport inequality, and multiply along the characteristic by its fixed scalar $N(0,b)^{-4}$. Only then let $\delta \downarrow 0$ and $\kappa \downarrow 0$.

The limiting quotient fields converge strongly in L^1 to θ in this cutoff limit: their difference from the prelimit $\mathcal{H}_{1,n}$ is uniformly small by joint uniform integrability. The signed work sources converge strongly in weighted L^1 by (5.5) and the cutoff identification following it. The shifted signed terms have a uniform exponent above one by (6.13). The denominator error is $O(\delta/\kappa)$. The retained absolutely continuous Fisher densities increase to m, so monotone convergence remains valid after multiplication by N^{-4}. The initial quotients converge to $\mathcal{H}_{1,0}/N(0,b)^4$. This proves the required upper inequality for θ/N^4 without a spatial commutator for raw θ or a raw product $u\theta$. Its complete source expression, along the characteristic, is

$$\begin{aligned} D_t(\theta/N^4) \le \frac{1}{N^4}\Big[&-4\int_B Mm + A : \tau(M\bar{s}) - \nu\mu \\ &- a\int_B MK : \langle \Pi(\log s + 1)\rangle\Big]. \end{aligned} \tag{7.4}$$

The raw conditional entropy source is $-4\int M|\nabla_q f_n|^2 + A_n : \tau(Ms_n) - a\int M(A_n : K)\log s_n$. The work term becomes the displayed mean work minus $\nu\mu$ by (5.9). Adding 1 inside the last logarithmic bracket changes its integral by zero, since $\int_B MK = \mathrm{Id}$ and $\operatorname{tr} A = 0$. Equation (7.4) is obtained only after the bounded-quotient normalization and ordered cutoff limits above.

7.3 The four terms and their signs

Subtract the lower squared-transform inequality from the upper entropy inequality. Put

$$e = \frac{\eta}{N^4} = \frac{\theta}{N^4} - \int_B Mh^2, \qquad C_1 = \langle \lambda^2 \rangle - y\langle \lambda / \sqrt{\log \lambda^2} \rangle. \tag{7.5}$$

The square passage from (7.2) gives the complementary balance

$$\begin{aligned} D_t \int_B Mh^2 \geq \frac{1}{N^4} \Big[& -2\mathcal{D}_y + 2\int_B M(\langle \Pi G \rangle q) \cdot \nabla_q y \\ & + 2\int_B MyK : \langle \Pi[(s-a)\Theta'(s) - G] \rangle + 2\int_B MyV \Big]. \end{aligned} \tag{7.6}$$

For the mean flux, weighted integration by parts and $\operatorname{tr} A = 0$ give $2\int_B M(Ayq) \cdot \nabla_q y = \int_B M(A : K)y^2$. Also $2L = y + \langle \lambda / \sqrt{\log \lambda^2} \rangle$, so $\overline{s} - 2yL + y^2 = C_1$. Insert $\langle \Pi G \rangle = Ay + \Gamma$ and $\langle \Pi s \Theta'(s) \rangle = AL + \Gamma'$ in (7.6), and subtract it from (7.4). The mean, Fisher, shifted, and covariance terms are, respectively,

$$(\partial_t + u \cdot \nabla_x)e \leq -\frac{\mathcal{B}_1 + \mathcal{B}_2 + \mathcal{B}_3 + \mathcal{B}_4}{N^4}, \tag{7.7}$$

where, with $s = \lambda^2$ inside brackets,

$$\mathcal{B}_1 = \nu\mu - A : \tau(MC_1), \tag{7.8}$$

$$\mathcal{B}_2 = 4\int_B Mm + 2\int_B MyV - 2\mathcal{D}_y, \tag{7.9}$$

$$\mathcal{B}_3 = a\int_B MK : \langle \Pi(\log s + 1) - 2y\Pi\Theta'(s) \rangle, \tag{7.10}$$

$$\mathcal{B}_4 = 2\int_B M\{(\Gamma q) \cdot \nabla_q y + (\Gamma' - \Gamma) : K\, y\}. \tag{7.11}$$

In (7.8), $\tau(MC_1) = \int_B KMC_1$. The unnormalized expressions in (7.8)–(7.11) specify the integrands in the normalized weak identity (7.7); no raw transport equation for an unbounded η is asserted. The stress covariance is (5.9); the remaining terms are precisely the mean stress mismatch, configuration chain-rule dissipation, shift, and gradient-density covariance when the two equations are subtracted.

Both maps $\lambda \mapsto \lambda\sqrt{\log \lambda^2}$ and $\lambda \mapsto \lambda/\sqrt{\log \lambda^2}$ are increasing on the shifted range. Their covariance gives $0 \leq C_1 \leq V_f$ by Cauchy–Schwarz. Thus (6.6) and Young's inequality imply

$$\mathcal{B}_1 \geq \nu\mu - \tfrac{1}{8}\varpi - C(1 + |A|^2)\eta, \qquad \mathcal{B}_2 \geq 2\varpi. \tag{7.12}$$

The second assertion is (6.9).

For the shift, use $\log s + 1 = 2\Theta'(s)G$ and split $\Pi = A + (\Pi - A)$. The fluctuating part is at most $C\sqrt{\mu\eta}$ because Θ' is bounded on $s \geq a$ and $\int M/d^2 < \infty$. The mean-gradient part is bounded by $C|A| \int MV_f/d$. The scalar estimate behind this bound is

$$|[\Theta'(\lambda^2) - \Theta'((\lambda')^2)][\lambda\sqrt{\log \lambda^2} - \lambda'\sqrt{\log(\lambda')^2}]| \leq C_a|\lambda - \lambda'|^2. \tag{7.13}$$

For $\lambda'/\lambda \leq 2$ it follows by the displayed scalar derivatives. For a larger ratio use $\Theta'(\lambda^2) \leq C\sqrt{\log \lambda^2}/\lambda$ and $|\lambda' - \lambda| \geq \lambda'/2$; the remaining factor is bounded by $C \log(\lambda^2)/\lambda^2$. Symmetry covers the reversed ratio. Therefore

$$|\mathcal{B}_3| \leq \tfrac{\nu}{8}\mu + \tfrac{1}{8}\varpi + C(1 + |A|^2)\eta. \tag{7.14}$$

Finally (6.11), Cauchy–Schwarz in q, and Hardy for y give

$$|\mathcal{B}_4| \leq \tfrac{\nu}{8}\mu + C\eta(\mathcal{D}_y + N). \tag{7.15}$$

All constants may depend on ν, k, a, T and the datum, with no smallness condition on these data and no positive lower bound on ν beyond $\nu > 0$.

7.4 Characteristic closure

Combining (7.7)–(7.15) and dropping the remaining favorable terms gives

$$(\partial_t + u \cdot \nabla_x)e \le C(1 + |A|^2 + N + \mathcal{D}_y)e, \qquad 0 \le e \le C/N^3. \tag{7.16}$$

The right side is globally L^1: use $N \ge 1$, $A \in L^2$, and $\mathcal{D}_y/N \in L^1$. Along almost every limiting characteristic, N is a finite constant, $\int |A|^2 < \infty$, and $\int \mathcal{D}_y/N < \infty$ by volume preservation. The coefficient of e is therefore integrable in time on that characteristic. The upper entropy and lower transform initial traces give $e(0) \le 0$, hence $e(0) = 0$. Applying the bounded mild transport inequality and the ordinary characteristic Gronwall inequality yields

$$\eta = 0 \quad \text{on every finite interval.} \tag{7.17}$$

This closes the conditional density fluctuation without any strong-gradient premise.

8 Strong density convergence and the original weak system

The definition of η, its vanishing, and uniform integrability of the conditional G_n^2 give

$$\int_0^T \int_{\mathbb{T}}^3 \int_B M\mathbb{E}|G_n^\omega - y|^2 \longrightarrow 0. \tag{8.1}$$

On $s, t \ge a$ the inverse transform satisfies $|s - t| \le C(1 + \Theta(s) + \Theta(t))|\Theta(s) - \Theta(t)|$. Cauchy–Schwarz thus turns (8.1) into strong L^1 convergence of the conditional densities to $M(\Theta^{-1}(y) - a)$. Their averages converge strongly to the same density, which must be the weak limit ψ. In particular every bounded kinetic test moment converges strongly in $L^2_{t,x}$: its difference is bounded by a constant times the conditional total variation, itself at most two, and tends to zero in L^1. Pairing with $A_n \rightharpoonup A$ in L^2 proves convergence of the exact configuration drag.

Strong density convergence identifies stress truncations strongly; the uniform collars (2.3) then give (1.9). Spatial transport passes as in Section 2. Moreover

$$\int M|\sqrt{\psi_n/M} - \sqrt{\psi/M}|^2 \le \int |\psi_n - \psi| \to 0.$$

The configuration gradients of these square roots are weakly $L^2(M)$, so their products identify the full diffusion flux

$$2M\sqrt{\psi_n/M}\,\nabla_q\sqrt{\psi_n/M} \longrightarrow M\nabla_q(\psi/M)$$

against bounded smooth tests up to the closed ball. The zero-flux variational identity therefore passes with its original boundary meaning. The spatial diffusion term vanishes at order ε_n by normalization.

Fluid convection passes by strong L^2 velocity convergence, viscosity by weak H^1 convergence, and forcing by strong stress convergence. Recover periodic pressure from the limiting divergence equation. The common initial terms pass in the time-integrated weak equations. Smooth test pairings are continuous in time by the flux bounds. Entropy uniform integrability extends this to weak L^1 density continuity: approximate a bounded measurable test by bounded smooth functions outside a set of arbitrarily small measure, then use the common uniform integrability on that set. The velocity energy class gives its weak L^2 time representative. Bounded $\log g_0$ and the initial entropy inequality yield the strong initial density trace by the relative entropy argument in Section 3.3.

Take an almost-everywhere-time subsequence of the strong convergences and use convex lower semicontinuity of entropy, viscosity, and configuration Fisher dissipation in (1.6). Drop its nonnegative spatial Fisher term. This gives the usual limiting initial-time inequality. If the positive family was selected to have almost-everywhere time-shifted inequalities, the higher conditional density moment makes the entropy densities uniformly integrable, and those inequalities pass too; that optional selection is not needed above. Finally use nested subsequences on integer time intervals and diagonalize. They define one global pair and one decreasing $\varepsilon_j \to 0$ with all the conclusions of Theorem 1.1. The fluid viscosity, FENE potential, full velocity-gradient drag, and zero configuration-flux condition have retained their original values and meanings throughout the limit.

AI use and materials. OpenAI Codex assisted with the argument, source checks, and mathematical exposition. Separate fresh Codex contexts supplied an AI reader trace and an AI referee report on the preceding draft; this revision responds to those reports. Independent expert review is requested. The manuscript contains the complete submitted analytic argument and uses no proof-critical numerical computation. Its editable LaTeX source accompanies the submission.

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